Image-Based Safety and Accessibility Risk Analysis Using Vision–Language Models

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Abstract

Safety and accessibility inspections are essential in public, workplace, and residential environments, yet they are often manual, subjective, and time-consuming. This project presents an image-based safety and accessibility analysis system that leverages a vision–language model to generate scene descriptions and structured hazard assessments from a single image. The system combines image captioning, rule-based scene classification, and deterministic hazard templates to produce interpretable and consistent safety reports. Rather than replacing human inspectors, the system is designed as a decision-support tool that highlights potential risks and accessibility concerns in a transparent and explainable manner. Experimental demonstrations show that the system can reasonably identify common hazards across diverse environments such as workspaces, streets, beaches, and shared pedestrian areas.

1. Introduction

There are many safety and accessibility hazards in public and professional settings that may not be immediately obvious, especially for people with mobility issues, vision impairments, or other accessibility requirements. Conventional safety inspections are usually carried out by hand, requiring skilled workers to visually evaluate surroundings and record risks. Although efficient, this procedure takes a lot of time and can change based on the inspector's judgement and experience.

Semantic information may now be extracted from images in natural language thanks to recent developments in computer vision and vision-language models. By automatically indicating possible issues from visual input, these models provide a chance to support safety and accessibility inspections. Nevertheless, a lot of current AI-based systems function as "black boxes," generating results that are challenging to decipher or audit.

By creating an image-based safety and accessibility analysis system that gives interpretability, determinism, and practical usability top priority, this research seeks to overcome these issues. The system doesn't try to make decisions about safety that are authoritative. Rather, it offers structured, comprehensible observations that can help ordinary users, accessibility auditors, and inspectors detect any hazards.

2. Problem Definition and Objectives

2.1 Problem Definition

This project focuses on extracting **safety hazards** and **accessibility barriers** from a single photo of a real-world environment (e.g., rooms, hallways, workspaces).
For any input image, the system produces three outputs:

- a short caption summarizing what is visible,
- a simple scene label based on transparent rules,
- a structured checklist-style report describing likely hazards and accessibility issues that could matter to a person moving through the space.

The goal is not perfect detection, but **consistent, explainable reporting** from visual information.

2.2 Objectives

The project aims to:

- Generate a clear, human-readable caption using a pretrained vision–language model.
- Assign a high-level scene category using rule-based keyword matching, so the logic is easy to inspect and modify.
- Convert the caption + scene label into a structured safety/accessibility report using deterministic templates for reliability.
- Keep the output stable and interpretable so a user can understand *why* a risk was flagged and what action is recommended.

3. System Architecture

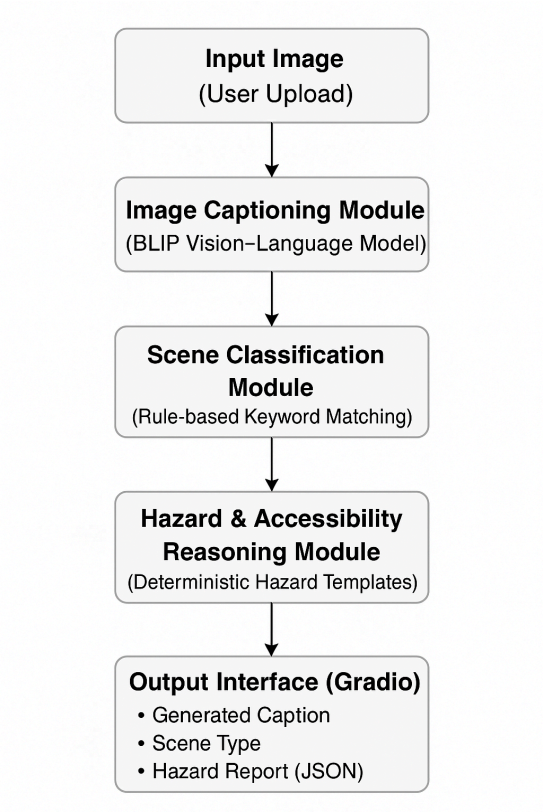

Figure 1. System architecture for the image-based safety and accessibility analysis pipeline.

The system follows a modular pipeline architecture consisting of the following components:

1. **Image Input Module**
   Accepts an image uploaded by the user through a graphical interface.
2. **Image Captioning Module**
   Uses a pretrained vision–language model to generate a textual description of the image.
3. **Scene Classification Module**
   Applies rule-based keyword matching on the generated caption to determine the scene type.
4. **Hazard and Accessibility Reasoning Module**
   Selects a deterministic hazard template corresponding to the identified scene.
5. **Output Interface**
   Displays the generated caption, detected scene type, and structured hazard report.

This modular design improves interpretability and allows each component to be analyzed independently.

4. Methodology

4.1 Image Captioning

Image understanding is performed using the BLIP (Bootstrapped Language–Image Pretraining) model. BLIP is a vision–language model capable of generating descriptive captions from images. The model is used as an intermediate semantic representation rather than a direct decision-maker.

The generated caption serves as a bridge between raw visual input and higher-level reasoning. GPU acceleration is used to ensure efficient inference.

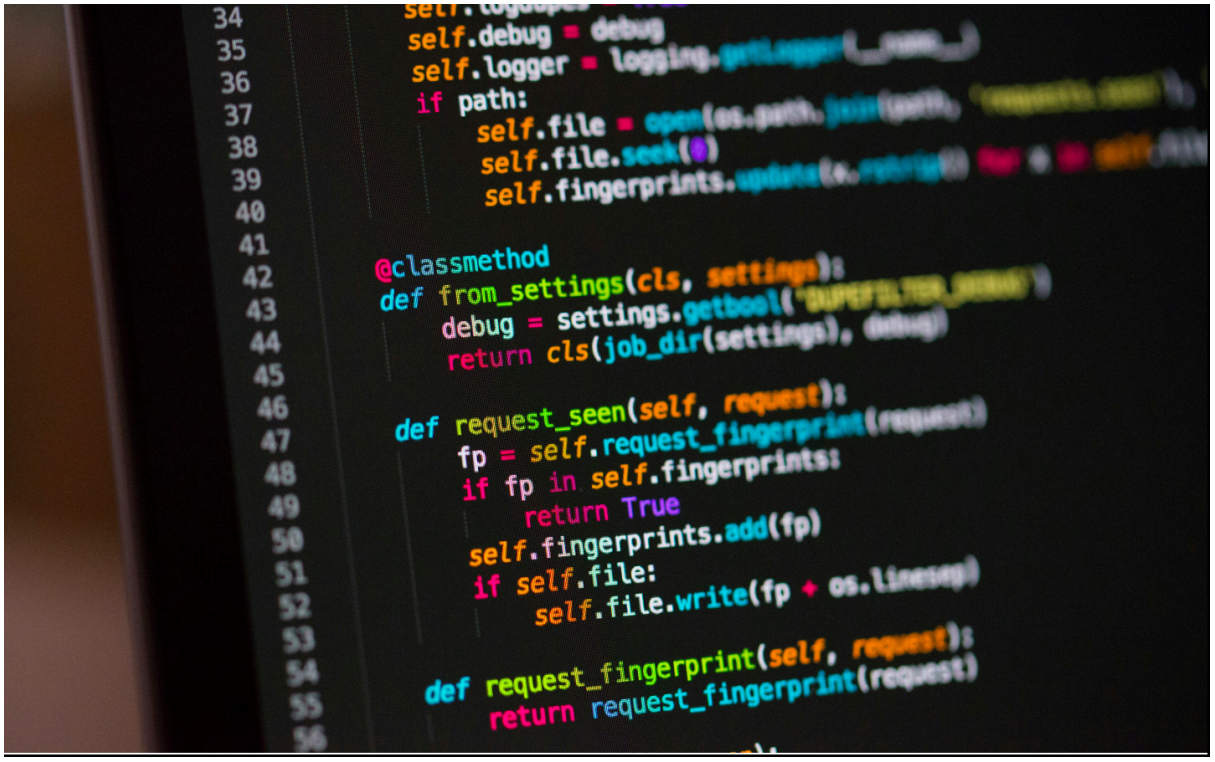

Figure 2. Example input image and generated caption produced by the image captioning module.

For the image shown in Figure 2, the image captioning module generated the caption: *"a computer screen with a bunch of code code."* This caption is used as an intermediate semantic representation of the visual content and is not treated as a final decision. Instead, it provides contextual information that is later processed by the scene classification and hazard analysis modules.

4.2 Scene Classification

Instead of training a learned scene classifier, this project uses a rule-based scene classification approach. After an image caption is generated by the vision–language model, the caption text is converted to lowercase and analyzed using a predefined set of keywords. Based on the presence of these keywords, the image is assigned to a high-level scene category such as workspace, street, beach, scooter area, or park.

This design choice is intentional and supports the project's emphasis on interpretability rather than predictive complexity. In particular, the rule-based approach ensures:

- Deterministic behavior, where the same input consistently produces the same scene label
- Ease of debugging, since classification decisions can be traced directly to keyword matches
- Transparency in decision-making, allowing the logic to be easily inspected and modified

Although rule-based classification lacks the adaptability and scalability of learned classifiers, it is well-suited for this project's goals. The approach prioritizes explainability and consistency over maximizing classification accuracy.

Instead of illustrating this process with a figure, the caption-to-scene mapping is described directly in text to reflect the simplicity of the underlying logic. Once a caption is generated, it is scanned for scene-specific keywords. If a match is found, the corresponding scene label is assigned deterministically. This text-based explanation avoids unnecessary visual abstraction while accurately representing the system's behavior.

4.3 Hazard and Accessibility Reasoning

Once a scene is identified, the system applies a predefined hazard and accessibility template corresponding to that scene. Each template produces a structured JSON output containing the following categories:

- Safety Hazards
- Accessibility Risks
- Tripping or Fall Dangers
- Wheelchair Accessibility Issues
- Obstacles or Narrow Pathways
- Environmental Conditions
- Corrective Actions

This template-based approach avoids hallucinated outputs and ensures consistent reporting. The system does not infer hazards beyond the scope of the selected template, reducing the risk of misleading conclusions.

```
··· Detected scene type: workspace
    {
      "Safety Hazards": [
        "Prolonged computer use may cause eye strain or musculoskeletal discomfort.",
        "Hot beverages or loose objects on the desk may pose minor contact hazards."
      ],
      "Accessibility Risks": [
        "Devices placed far back on the desk may be difficult for some users to reach.",
        "Low visual contrast on the desk surface may hinder visibility for users with low vision."
      ],
      "Tripping or Fall Dangers": [
        "Cables near the desk edge could fall and clutter the floor.",
        "Objects positioned near the edge may fall and become tripping hazards."
      ],
      "Wheelchair Accessibility Issues": [
        "Desk height and under-desk space may not allow comfortable access for wheelchair users.",
        "Limited leg clearance may restrict positioning for mobility devices."
      ],
      "Obstacles or Narrow Pathways": [
        "Cluttered desk items may restrict arm movement.",
        "Peripherals and desk accessories may reduce usable workspace."
      ],
      "Environmental Conditions": [
        "Glare from overhead lighting or monitors may affect comfort.",
        "Room temperature or noise levels may influence usability."
      ],
      "Corrective Actions": [
        "Organize desk items to keep surfaces clear and reduce clutter.",
        "Adjust ergonomics, lighting, and seating to improve comfort and safety."
      ]
    }
```

Figure 3. Example structured hazard and accessibility report generated for a detected workspace scene.

5. Implementation Details

The system is implemented using Python and PyTorch. The BLIP model is loaded using the Hugging Face Transformers library and executed on a CUDA-enabled GPU. Image preprocessing is handled using the Pillow library.

A Gradio-based user interface is implemented to allow users to upload images and view results interactively. The interface displays the generated caption, detected scene type, and structured hazard report in real time.

The modular structure of the codebase allows for future extensions such as additional scene categories or alternative captioning models.

6. User Interface and Demonstration

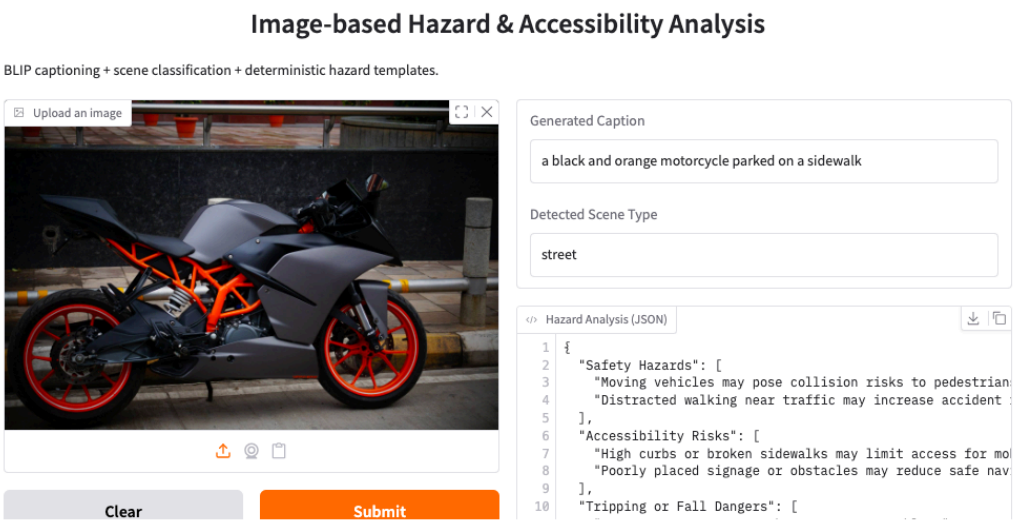

Figure 4. User interface demonstrating image upload, generated caption, detected scene type, and structured hazard and accessibility analysis.

The user interface enables non-technical users to interact with the system. Users upload an image, after which the system automatically generates outputs without requiring manual configuration.

The interface design emphasizes clarity by separating descriptive output from structured hazard analysis. This makes the system suitable for demonstrations, educational use, and preliminary inspections.

7. Evaluation and Limitations

7.1 Evaluation

This project primarily uses qualitative evaluation. The system was tested on a variety of images representing different environments, including workspaces, outdoor scenes, and public areas. In most cases, the system produced reasonable captions, correctly identified the scene type, and generated appropriate hazard templates.

7.2 Limitations

Despite its strengths, the system has several limitations:

- Scene classification relies on keyword matching and may fail for ambiguous captions
- Hazard outputs are template-based and not image-specific beyond scene type
- The system does not provide quantitative risk scores
- Static images limit temporal understanding of hazards

These limitations are acknowledged as trade-offs for interpretability and determinism.

8. Ethical Considerations

The system is designed as a support tool rather than an authoritative decision-maker. Incorrect or incomplete hazard identification could lead to over-reliance if used improperly. To mitigate this risk, outputs are intentionally conservative and descriptive.

Bias in image interpretation and dataset limitations may also affect results. Users are encouraged to treat system outputs as guidance rather than final judgments.

9. Future Work

Future extensions of this project may include:

- Video-based hazard analysis
- Learned scene classification models
- Integration with accessibility standards and building codes
- Mobile deployment for field inspections
- Multilingual output support

10. Conclusion

This project demonstrates the feasibility of using vision–language models for image-based safety and accessibility analysis while maintaining transparency and interpretability. By combining image captioning, rule-based reasoning, and deterministic hazard templates, the system provides structured insights that can assist human inspectors. The project highlights the importance of explainable AI in safety-critical domains and lays the groundwork for future enhancements.